

8. Suppose you are training your reward model using Bradley-terry model. You have two preference data points in your batch. For the first data point, the transformer model gives a score of 3.9 to 'prompt, winning sample', and a score of 4.5 to 'prompt, losing sample'. For the second data point, the transformer model gives a score of 3.3 to 'prompt, winning sample' and 2.7 to 'prompt, losing sample'. What will be the corresponding loss for this batch? [3.5]
9. Suppose you want to use T5 model, which has a model dimensionality of 1024 and 24 layers, for the tasks of: (1). text summarization, (2). question answering and (3). text simplification. You can either use full fine-tuning for each task or prompt-tuning (soft-prompts of length 10 for each task). How many extra parameters will you have to store in both of these scenarios at inference time, apart from the T5 parameters? [4]
10. Suppose you have a vocabulary with 3 tokens, '{car, jeep, petrol}'. At inference time in the decoder, suppose that the logits are '{3, 0, 1}'. Which of the following decoding strategies can you use so that the probability of selecting the word 'jeep' is less than 0.02, while still allowing you to select the other two words?
- Greedy decoding
 - Random sampling with temperature
 - Top-k sampling
 - Top-p sampling

For the valid sampling methods, please also provide the required hyper-parameter values (e.g., value of k in top-k). [3.5]

11. Suppose you are training a transformer encoder-decoder for the task of machine translation from source 'English' to target 'Bengali'. Assume that the source vocabulary is 30k, while the target vocabulary is 40k. Across encoder and decoder, you are using model-dimensionality of 1,024 and the feed-forward hidden layer has 2,048 units. Assume that there are 16 attention heads for all kinds-of-attention, as per the default setup. Also, assume that both encoder and decoder use the (same) trainable positional embeddings, and the maximum sequence length is 512.
- However, while the encoder uses 24 layers, you only use 12 layers for the decoder. How many parameters does the model has? Assume that there is weight tying for 'unembedding'.
- Now, you make a variation where you use a single key and value across all 16 attention heads (while using 16 different 'queries', one for each head). What will be the number of parameters? [7]

Instead of training reward model, DPO derives an analytic expression for the reward model.

$$RM_{\theta}(x, \hat{y}) = \beta \log \frac{p_{\theta}^{RL}(\hat{y}|x)}{p^{PT}(\hat{y}|x)} + \beta \log Z(x)$$

How is this used to derive the final DPO loss?

5. Low-rank Adaptation, Quantization

- (a) Suppose that you want to use Low-rank adaptation methods to train the GPT-2 model, which has a model dimensionality of 1600, 48 decoder layers and 25 attention heads, for the tasks of text simplification. Consider the following scenarios. What will be the number of parameters to be trained? What will be the number of parameters at the inference time?
- You are using LoRA only for the query and key matrices, each with a rank of 8.
 - You are using 2 adapters per layer, each with a down projection of 8.

[3+3 = 6]

- (b) Suppose that you are using QLoRA to achieve 2-bit quantization with values: $\{-1, -0.3, 0.5, 1\}$. Assume your input tensor is $[38, -3, -50, 25]$. What will you need to store in the quantized format. Also, what will be the output after dequantization? [3]
- (c) Suppose that you are using double quantization for a 10×10 weight matrix containing FP32 values to store as Int8. You are using a block size of 20 for the double quantization. How many bits would you save? [3]